Sec. VI-E: hand-calculable bounds

numbering continues from (105)

Eq. (105) bounds the shifted critical moment by the two channel maxima, and (93) supplies the constants that multiply them. What remains is to evaluate $\rho_{\varphi,\max}$ and $\rho_{GE,\max}$ themselves. Both are collected here rather than at the point where each channel was introduced, so that the arithmetic appears once.

Nothing below requires a run. The weight comes from a scale, the pivot arm and z_{CoM} from the box of Sec. VI-D, the inertia from CAD, the tilt cap and the ramp rates from the excitation protocol of Sec. V-B, and the ground-effect slope from table 3. The bound is therefore a statement made before the campaign, not a description of it.

The window, which only the coupling channel needs. The nominal excursion of (20) is $\varphi_{\text{end}} = \big(\dot{M}/Wz_{CoM}C_2\big)\Lambda(x)$, and every term of Λ is positive, so $\Lambda(x) \geq x^3/6$. Imposing the tilt cap $\varphi_{\text{end}} \leq \varphi_{\max}$ therefore inverts in closed form, and with $J_P = Wz_{CoM}/C_2^2$ the window length and the moment increment across it follow without solving for x:

$$\tau_{\text{end}} \leq \left(\frac{6\varphi_{\max}J_P}{\dot{M}}\right)^{1/3}, \qquad \Delta M_{\text{win}} = \dot{M}\tau_{\text{end}} \leq \big(6\varphi_{\max}J_P\dot{M}^2\big)^{1/3}. \tag{106}$$

The inertia enters at its upper end, $J_P \leq J_{CoM}^{\text{CAD}} + m\big(z_{CoM}^2 + \big(l_p + p_{\text{off}}\big)^2\big)$, which is the unfavourable side here: (82) discards that CAD term, but retaining it is what makes (106) a bound rather than an estimate.

The gravity channel. By (85) the remainder is $\rho_\varphi = \frac{1}{2}G''(\varphi^*)\varphi^2$ with $G''(\varphi^*) = W\big(l_p + p_{\text{off}}\big)\cos\varphi^* - Wz_{CoM}\sin\varphi^*$, which decreases on $[0, \varphi_{\max}]$ and is therefore largest at $\varphi^* = 0$. Hence

$$\rho_{\varphi,\max} \;\leq\; \tfrac{1}{2}\,W\big(l_p + p_{\text{off}}\big)\,\varphi_{\max}^2, \tag{107}$$

evaluated at the ballasted weight and at the widest arm the offset box admits. It does not depend on the ramp rate: the tilt cap alone fixes it.

The coupling channel. Term (iv) of (80) is $\rho_{GE}(\tau,\varphi) = \beta_M\dot{M}\tau\varphi$, so its maximum over the window is the product of the moment increment and the excursion, both already bounded:

$$\rho_{GE,\max} = |\beta_M|\,\Delta M_{\text{win}}\,\varphi_{\text{end}} \;\leq\; |\beta_M|\,\varphi_{\max}\big(6\varphi_{\max}J_P\dot{M}^2\big)^{1/3}. \tag{108}$$

This is the only place the ramp rate enters, and it enters as $\dot{M}^{2/3}$.

The bound, in six lines. Substituting (107) and (108) into (105) and dividing by the unloaded weight gives the equivalent offset error. No hyperbolic function appears, C_2 never enters, and x is never solved for — $\frac{1}{7}$ and $\frac{1}{5}$ in (93) are suprema over *all* x:

$$\boxed{\big|\Delta M_{\text{crit},e}\big| \leq \frac{\rho_{\varphi,\max}}{7} + \frac{\rho_{GE,\max}}{5}, \qquad \big|\Delta\lambda_{\text{off}}\big| \leq \frac{1}{W}\left[\frac{\rho_{\varphi,\max}}{7} + \frac{\rho_{GE,\max}}{5}\right].} \tag{109}$$

Over the admissible box, with $W = 31.59$ N, $z_{CoM} = 0.30$ m, $J_{CoM}^{\text{CAD}} = 0.0537$ kg m^2, $\varphi_{\max} = 10°$ and the arms and slopes above:

axis	$\dot{M}$ [N m/s]	J_P [kg m^2]	ΔM_{win} [N m]	$\rho_{\varphi,\max}$ [mN m]	$\rho_{GE,\max}$ [mN m]	$\lvert\Delta\lambda_{\text{off}}\rvert$ [mm]
roll	0.10	0.4260	0.165	76.98	0.99	0.372
	0.45	0.4260	0.449	76.98	2.70	0.384
	1.20	0.4260	0.863	76.98	5.19	0.400
pitch	0.10	0.3979	0.161	62.55	0.72	0.302
	0.45	0.3979	0.439	62.55	1.97	0.310
	1.20	0.3979	0.843	62.55	3.79	0.322

The same row, by hand. Roll at $\dot{M} = 1.20$ N m/s, every intermediate carried so the row can be checked on a calculator:

	quantity	value
	$\varphi_{\max} = 10^\circ$	0.174533 rad
	$m = W/g = 31.59/9.81$	3.220183 kg
	$z_{CoM}^2 + \left(l_p + p_{\text{off}}\right)^2 = 0.09 + 0.0256$	0.115600 m^2
	$m\left(z_{CoM}^2 + \left(l_p + p_{\text{off}}\right)^2\right)$	0.372253 $\mathrm{kg\,m}^2$
(1)	$J_P \leq 0.0537 + 0.372253$	0.425953 $\mathrm{kg\,m}^2$
	$6\varphi_{\max} J_P \dot{M}^2 = 6(0.174533)(0.425953)(1.44)$	0.642322
(2)	$\Delta M_{\text{win}} \leq (0.642322)^{1/3}$	0.862815 N m
	$\varphi_{\max}^2$	0.030462
(3)	$\rho_{\varphi,\max} = \frac{1}{2}(31.59)(0.160)(0.030462)$	0.076983 N m
(4)	$\rho_{GE,\max} = (0.03446)(0.862815)(0.174533)$	0.005189 N m
	$\rho_{\varphi,\max}/7$	10.998 mN m
	$\rho_{GE,\max}/5$	1.038 mN m
(5)	$\lvert\Delta M_{\text{crit},e}\rvert \leq$ sum	12.035 mN m
(6)	$\div W = 30.08$ N	0.400 mm

Note which weight goes where: $W = 31.59$ N, the ballasted value, enters (3) because a heavier vehicle makes the gravity remainder larger, while $W = 30.08$ N, the unloaded value, enters (6) because a lighter vehicle turns a given moment error into a larger offset. Each is taken at whichever end of the range is unfavourable for the quantity it appears in.

The worst admissible case is 0.400 mm for roll and 0.322 mm for pitch, of which the gravity channel supplies 91 and 92%. Two readings follow. The inertia, the one input that is not directly measured, reaches the result only through $\rho_{GE,\max}$ and only as a cube root: halving J_P moves the bound by -1.8% and doubling it by $+2.2\%$, so a coarse CAD figure suffices. And precision is bought at the tilt cap rather than at the ramp: (107) is quadratic in $\varphi_{\max}$ while (108) is only $\dot{M}^{2/3}$, so a twelve-fold reduction in ramp rate moves the bound by 7% whereas halving the cap quarters it. The 10° used here was chosen for onset visibility on the lightly loaded configurations; at 5° the worst case would fall to 0.105 mm for roll and 0.084 mm for pitch — a little short of a factor of four, because $\rho_{GE,\max}$ carries only $\varphi_{\max}^{4/3}$ — at the price of a 13 to 17% shorter fitting window and a rate at the cap that drops from 0.92 to 0.48 rad/s on the slowest ramp, roughly halving the signal the onset fit has to work with.

The angular rate error. Eq. (90) bounds the rate deviation itself, and unlike (109) it needs the window — not a bound on the window, but the window itself. The distinction matters. In (109) the reduction factors were replaced by their suprema $\frac{1}{7}$ and $\frac{1}{5}$, which hold for every x, so no value of x had to be committed to. Here the kernel gain $\sinh x$ survives, and the product $R(x)\sinh x$ is what the bound depends on; substituting a mere upper bound $\bar{x}$ would require knowing that this product is monotone. It is, but nothing needs to rest on that, because x is not free: the run ends when the tilt cap is reached, so by (20)

$$\Lambda(x) \;=\; \frac{\varphi_{\max} W z_{CoM} C_2}{\dot{M}}, \qquad \Lambda(x) = \sinh x - x, \tag{110}$$

a scalar equation with a unique root, since Λ is strictly increasing from 0. The cubic shortcut of (106) is the same relation with $\Lambda(x)$ replaced by $x^3/6$; it is what makes (106) closed-form, and it costs little there because ΔM_{win} enters (109) only through the 8–9% coupling channel. It would cost a great deal here, where Λ exceeds $x^3/6$ by a factor of 3.6 at the slowest ramp and the difference is then exponentiated by sinh. So (110) is solved as it stands.

The reduction factors themselves are elementary integrals, and both windows close in finitely many terms:

$$\begin{aligned} R_\varphi(x) &= \frac{A_\varphi(x)}{x\Lambda(x)^2}, \qquad A_\varphi(x) = \tfrac{1}{4}\sinh 2x - \tfrac{1}{2}x - 2x\cosh x + 2\sinh x + \tfrac{1}{3}x^3, \\ R_{GE}(x) &= \frac{A_{GE}(x)}{x^2\Lambda(x)}, \quad A_{GE}(x) = x\cosh x - \sinh x - \tfrac{1}{3}x^3. \end{aligned} \tag{111}$$

Finally C_2 has a geometric ceiling in which the mass cancels,

$$C_2^2 = \frac{W z_{CoM}}{J_P} \;\leq\; \frac{W z_{CoM}}{m\left(z_{CoM}^2 + \left(l_p + p_{\text{off}}\right)^2\right)} = \frac{g\, z_{CoM}}{z_{CoM}^2 + \left(l_p + p_{\text{off}}\right)^2}, \tag{112}$$

giving 5.05 s^{-1} for roll and 5.25 for pitch. Note that here the inertia is taken at its *lower* end, $J_P = m(z_{CoM}^2 + (l_p + p_{\text{off}})^2)$, the opposite of (106): $J_P C_2 = \sqrt{J_P W z_{CoM}}$ sits in the denominator of (90), and a smaller J_P both enlarges that quotient and raises C_2, hence x. The two ends coexist without contradiction because they enter different factors — $\rho_{GE,\max}$ keeps the upper end it was given at line (4), the kernel takes the lower — and each is the end unfavourable to the factor it appears in. Continuing the roll row above:

	quantity	value
	$J_P = m\big(z_{CoM}^2 + (l_p + p_{\text{off}})^2\big)$, lower end	0.372253 kg m^2
	$W z_{CoM} = (31.59)(0.30)$	9.477000 N m
(7)	$C_2 = \sqrt{W z_{CoM}/J_P}$, and (112) agrees	5.045639 s^{-1}
	$\varphi_{\max} W z_{CoM} C_2/\dot{M} = (0.174533)(9.477)(5.045639)/1.20$	6.954777
(8)	x, the root of $\sinh x - x = 6.954777$	2.993015
	check: $\sinh(2.993015) = 9.947796$, less x	6.954781
	$\Lambda(x) = 6.954781$, $\Lambda(x)^2$	48.368980
	A_φ from (111), $\frac{1}{4}(198.914782) - \frac{1}{2}(2.993015)$	
	$- 2(2.993015)(9.997932) + 2(9.947796) + 8.937281$	17.217140
(9)	$R_\varphi = 17.217140/\big[(2.993015)(48.368980)\big]$	0.118928
	$A_{GE} = (2.993015)(9.997932) - 9.947796 - 8.937281$	11.038884
(10)	$R_{GE} = 11.038884/\big[(2.993015)^2(6.954781)\big]$	0.177184
	$R_\varphi\, \rho_{\varphi,\max} = (0.118928)(0.076983)$	0.009155
	$R_{GE}\, \rho_{GE,\max} = (0.177184)(0.005189)$	0.000919
(11)	$\bar{\rho} = R_\varphi \rho_{\varphi,\max} + R_{GE} \rho_{GE,\max}$	0.010075 N m
	$J_P C_2 = \sqrt{J_P W z_{CoM}}$	1.878255
(12)	$\lvert e_\omega(\tau_{\text{end}})\rvert \le \bar{\rho} \sinh x/(J_P C_2)$	0.053360 rad/s
	$C_1 = \dot{M}/W z_{CoM}$	0.126622
	$\cosh x - 1$	8.997932
(13)	$\omega_{\text{nom,end}} = C_1(\cosh x - 1)$	1.139339 rad/s
(14)	relative, (12) $\div$ (13)	4.68%

Line (11) is where this calculation parts company with (109). There the average was bounded by $\rho_{\varphi,\max}/7 + \rho_{GE,\max}/5$, the $x \to 0$ suprema; here the same average is evaluated at the x the run actually reaches, which is smaller — 0.010075 against 0.012035 — because R_φ and R_{GE} decrease. The two are consistent, the second simply being the sharper of the pair. Across the ramp range the roll axis gives

$\dot{M}$ [N m/s]	x	$\bar{\rho}$ [mN m]	$\lvert e_\omega\rvert$ [rad/s]	$\omega_{\text{nom,end}}$ [rad/s]	relative
0.10	5.18	7.041	0.332	0.925	35.9%
0.45	3.80	8.751	0.104	1.015	10.3%
1.20	2.99	10.075	0.053	1.139	4.7%

Pitch is milder on all three rows, 30.1, 8.6 and 3.9%, the shorter pivot arm reducing $\rho_{\varphi,\max}$ faster than it raises C_2.

Two things are worth reading off, and the first is not the window. It is natural to read the slow rows as the forcing having had longer to act on an unstable plant, and for the *absolute* deviation that is exactly right: 0.053 grows to 0.332 rad/s because $\sinh x$ grows ninefold. But ω_{nom} is the output of the same plant, driven by the ramp instead of by ρ, so the amplification is common to both and cancels. Using $J_P C_2^2 = W z_{CoM}$ the ratio collapses to

$$\frac{\lvert e_\omega\rvert_{\text{end}}}{\omega_{\text{nom,end}}} \le \coth\frac{x}{2} \cdot \frac{\bar{\rho}\, C_2}{\dot{M}}, \tag{113}$$

in which the window survives only through $\coth(x/2)$, running from 1.106 to 1.011 across the sweep — eight per cent, and in the favourable direction. The realised windows are shorter than these rows assume and so make that term larger, 1.258 down to 1.038, but twenty-six per cent in place of eight leaves the reading unchanged. What remains is the disturbance measured against $\dot{M}/C_2$, the moment the ramp sweeps in one time constant $1/C_2 = 0.198$ s of the error dynamics: 238 mN m at the fastest ramp against 19.8 at the slowest, while $\bar{\rho}$ only falls from 10.1 to 7.0. Twelve times slower is twelve times worse, and almost nothing else is happening. Precision here is therefore bought by ramping faster, the opposite of the rule for (109), which is why the two bounds want different operating points and why the section reports both.

And these percentages are relative to the peak nominal rate, so they overstate what the estimator suffers. The onset is located by the normalised correlation of (107), not by the rate at a single instant, so the quantity that reaches the offset budget is (109), not this. The rate bound is reported because it is the one a reader can compare against a gyro specification, not because the budget rests on it.

What the measured residual does, and why it differs. A reader who puts this table beside the fit residual of Sec. VIII will find two mismatches, and both are informative rather than symptoms of a broken bound. The residual is 7.2% of the peak in the median and *flat* across the ramp sweep, so it neither falls as the table predicts nor stays under the 4.7% row at the fastest ramp.

The first mismatch is the window. The rows above are set at $\varphi_{\max} = 10°$, whereas the runs stop at 4.8° in the median, and they stop for a different reason: the excitation window closes on the moment, not on the tilt. Taking x from each run's own fit — C_2 the configuration's calibrated value, τ_{end} the post-onset span it covered — gives $x = 3.97$ falling to 2.17 across the sweep on roll, against the 5.18 to 2.99 the tilt limit alone implies. Evaluating (110)–(112) at those realised windows, with the tilt each one reached, lowers the seven rows to 5.1, 2.9, 2.4, 2.4, 1.6, 1.4 and 1.1%. Most of that is the excursion: $\rho_{\varphi,\max}$ is quadratic in it. Against those the measured residual runs 5.1, 7.5, 6.5, 7.1, 3.9, 6.2 and 7.5% — level with the bound at the slowest ramp and above it thereafter.

The second is structural, and it is why the two were never the same quantity. Eq. (90) bounds the deviation of the measurement from the nominal response at the *true* onset with the *true* constants. A fit residual is the deviation from a curve whose onset was chosen and whose constants were calibrated. Subtracting,

$$\underbrace{y - \hat{\omega}(t;\hat{t}_{\mathrm{c}}) - C}_{\text{residual}} = \underbrace{e_{\omega}}_{(90)} - \underbrace{\delta t_{\mathrm{c}}\,\chi}_{\text{absorbed}} - \underbrace{(C-b)}_{(106)} - \underbrace{\delta C_1 \frac{\partial \omega}{\partial C_1} - \delta C_2 \frac{\partial \omega}{\partial C_2}}_{\text{absent from (90) by construction}}, \tag{114}$$

and the last pair exists only because the reference curve is calibrated rather than true. The excess of the measurement over the table is that pair, not a failure of the bound.

It is identifiable, moreover. If the residual were the modelled forcing, the runs assigned a larger bound would show a larger residual; within each ramp rate the rank correlation is −0.312, the wrong sign. Nor is the absorption of the fit hiding ρ: the estimator removes the component of the deviation that looks like a shifted onset, by (103), and that is the component concentrated *early*, whereas the Chebyshev step above rests on a ρ concentrated at the window's end — which the fit cannot absorb. What the residual does track is the pre-onset noise floor, +0.465 within a ramp rate, at 2.3 times its level and without scaling with the excursion: a disturbance floor that the tip raises but does not create. So (110)–(112) bound one channel, and in these experiments that channel sits below the floor at every rate — which is the outcome the section wants, since it means the forcing the estimator cannot absorb is not what limits the result. The measured budget, including the channels bounded nowhere here, is Sec. VIII-E.

What (109) covers. The displacement of the identified onset caused by the two forcing terms the estimator cannot absorb, and nothing else. The baseline error of (104), the departure of the applied moment from a linear ramp, and the ground-contact geometry all lie outside it. Those are measured rather than bounded a priori, and are reported with the realised budget in Sec. VIII-E, where the same runs give an actual propagated error some fifty times inside (109).